

Scalar Hermitian Perron Saturation and Cycle-Holonomy Defects: Exact Equality, Frustration Energies, and Weighted Bounds

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Abstract

Let P be a finite real symmetric nonnegative matrix with zero diagonal, and let $H = H^*$ be a scalar complex Hermitian matrix with zero diagonal and $|H_{ij}| \leq P_{ij}$. The familiar Perron majorization

$$\|H\| \leq \rho(|H|) \leq \rho(P)$$

does not identify when completion slack or complex phases make either inequality strict.

For connected P , let $Pr = \rho(P)r$ with $r_i > 0$, write $H_{ij} = q_{ij}\phi_{ij}$, and set $x_i = r_i f_i$. We prove the exact two-sector ground-state identity

$$x^*(\rho I - \sigma H)x = \sum_{\{i,j\}} r_i r_j [q_{ij}|f_i - \sigma \phi_{ij} f_j|^2 + (P_{ij} - q_{ij}) \cdot (|f_i|^2 + |f_j|^2)],$$

for $\sigma \in \{\pm 1\}$. Consequently,

$$\|H\| = \rho(P)$$

if and only if, for some sector σ , every amplitude bound is saturated and

$$H = \sigma D P D^*$$

for a diagonal unitary D . In the saturated case this switching condition is equivalent to

$$\text{Hol}_\phi(C) = \sigma^{|C|}$$

on every cycle. Only a fundamental-cycle basis must be checked. In particular, every scalar phase pattern on a tree is switchable and $\|H\| = \rho(|H|)$; a tree can never create phase-frustration saving.

We express the positive and negative spectral defects as exact Perron-weighted magnetic energies. Fundamental-cycle defects give explicit upper bounds, while any defective cycle gives a resistance type lower bound. The exact flux spectrum of C_n shows that small holonomy defects produce quadratic, not linear, spectral loss. We also give the reducible equality criterion and the exact decomposition

$$\rho(P) - \|H\| = [\rho(P) - \rho(|H|)] + [\rho(|H|) - \|H\|].$$

These results close the finite scalar Hermitian equality question in the dominated zero-diagonal setting. They neither show that the unarchived growing fixed-shift carrier has defective dominant cycles nor provide a mass-normalized exponent saving. The Hermitian row graph used here is distinct from the sparse completion incidence graph.

Keywords: Perron majorant; Hermitian matrix; magnetic graph; switching; cycle holonomy; frustration energy; operator norm; fixed shift.

Convention. Every support graph in the main theorems is the exact post-aggregation scalar Hermitian row graph. A physical specialization retains one fixed nonzero shift h_0 , literal coefficients, complete keys, exact support, and global raw normalization.

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1 Introduction

The fixed-shift completion hierarchy produces nonnegative matrices that majorize the absolute entries of a scalar Hermitian carrier [3]. Its operator estimate ends with

$$\|H\| \leq \rho(|H|) \leq \rho(P). \quad (1)$$

TPC-78 classifies certain forest components of the *completion incidence program* [4]. It does not answer when the Hermitian inequality (1) is strict, because the graph governing that question has physical rows as vertices and post-summation matrix entries as edges.

Two mechanisms can create slack. First, $|H_{ij}| < P_{ij}$ leaves amplitude or completion slack. Second, even when $P = |H|$, edge phases can be inconsistent around cycles. These are different phenomena, and they must be separated before a strict Perron comparison can be interpreted.

Main results. We derive an exact Perron ground-state transform that is simultaneously valid for the positive and negative extremal sectors. It yields a necessary-and-sufficient equality theorem:

$$\|H\| = \rho(P) \iff |H_\alpha| = P_\alpha, \quad H_\alpha = \sigma D_\alpha P_\alpha D_\alpha^*$$

on a Perron-dominant connected component, for some $\sigma \in \{\pm 1\}$. In the saturated connected case, diagonal switching is equivalent to a $U(1)$ cycle-holonomy condition.

The two sectors matter. Frustrating the positive Perron eigenvector does not force operator-norm strictness if the negative sector still saturates. A signed triangle with total phase -1 is the smallest example. Genuine phase saving requires both sectors to have a positive defect.

Quantitative content. The ground-state identity is an exact magnetic Dirichlet energy. It gives a variational formula, a fundamental-cycle upper certificate, and a defective-cycle lower certificate in terms of Perron conductances. The flux cycle C_n is solved in closed form and shows quadratic small-angle loss. Thus finite strictness alone cannot be upgraded to an asymptotic power saving.

Scope. The phase classification is scalar and Hermitian. A general real-linear completion block has no single $U(1)$ phase. A matrix-valued block can saturate on a common invariant subspace without having trivial matrix holonomy. A non-Hermitian carrier requires singular-vector or dilation arguments, while a nonsymmetric nonnegative majorant need not bound the Euclidean operator norm by its spectral radius.

Progress label. The identities are exact L0 spectral algebra and advance the L1 operator interface for a literal scalar Hermitian carrier. They do not provide the missing growing-carrier census or any fixed- h_0 arithmetic exponent estimate.

Organization. Section 2 defines the two-sector phases. Section 3 proves the exact sum of squares. Section 4 classifies saturation. Sections 5 and 6 give switching and tree consequences. Sections 7 and 8 prove weighted bounds and the cycle model. Section 9 handles reducible majorants. Sections 10 to 12 record the interface, counterexamples, and stop rules.

2 Dominated scalar Hermitian normal form

Let V be a finite index set. Let $P = (p_{ij})_{i,j \in V}$ be real symmetric, entrywise nonnegative, and zero on the diagonal. Let $H = H^* \in \mathbb{C}^{V \times V}$ be zero on the diagonal and satisfy

$$|H_{ij}| \leq p_{ij}. \quad (2)$$

Write

$$q_{ij} = |H_{ij}|, \quad H_{ij} = q_{ij} \phi_{ij}$$

on every edge with $q_{ij} > 0$, where

$$|\phi_{ij}| = 1, \quad \phi_{ji} = \overline{\phi_{ij}}.$$

The value of ϕ_{ij} on a zero q -edge is irrelevant.

Definition 2.1 (Row support graph). The undirected graph G_P has vertex set V and edge $\{i, j\}$ exactly when $p_{ij} > 0$. The graph G_q is defined similarly using $q_{ij} > 0$.

These are exact post-aggregation graphs. In particular, two atomic contributions that cancel in H_{ij} remove the edge from G_q , even if they were separately nonzero before summation.

Assume first that G_P is connected and $P \neq 0$. Perron–Frobenius theory [1, 2] gives

$$Pr = \rho r, \quad r_i > 0, \quad \sum_i r_i^2 = 1, \quad (3)$$

where $\rho = \rho(P)$.

Definition 2.2 (Two spectral sectors). For $\sigma \in \{+1, -1\}$, define

$$\delta_\sigma(P, H) = \rho(P) - \lambda_{\max}(\sigma H). \quad (4)$$

Thus

$$\delta_+ = \rho - \lambda_{\max}(H), \quad \delta_- = \rho + \lambda_{\min}(H),$$

and

$$\rho(P) - \|H\| = \min\{\delta_+, \delta_-\}. \quad (5)$$

The sign σ is global. It is not an independently chosen sign on each edge.

Definition 2.3 (Cycle holonomy). For an oriented cycle

$$C = (v_0, v_1, \dots, v_\ell = v_0)$$

contained in G_q , set

$$\text{Hol}_\phi(C) = \prod_{k=0}^{\ell-1} \phi_{v_k v_{k+1}}, \quad \text{Hol}_\sigma(C) = \sigma^\ell \text{Hol}_\phi(C). \quad (6)$$

Reversing orientation conjugates the holonomy, so the condition that it equal 1 is orientation independent.

3 The exact Perron ground-state transform

Theorem 3.1 (Dominated Hermitian sum of squares). *Assume G_P is connected. For $\sigma \in \{\pm 1\}$ and $x_i = r_i f_i$,*

$$\boxed{x^*(\rho I - \sigma H)x = \sum_{\{i,j\} \in E(G_P)} r_i r_j \left[q_{ij} |f_i - \sigma \phi_{ij} f_j|^2 + (p_{ij} - q_{ij})(|f_i|^2 + |f_j|^2) \right].} \quad (7)$$

On an edge with $q_{ij} = 0$, the first term is interpreted as zero.

Proof. For one undirected edge, the summand expands to

$$p_{ij} r_i r_j (|f_i|^2 + |f_j|^2) - 2\sigma q_{ij} r_i r_j \text{Re}(\overline{f_i} \phi_{ij} f_j).$$

Summing its diagonal part over edges and using (3) gives

$$\sum_i r_i |f_i|^2 \sum_j p_{ij} r_j = \rho \sum_i r_i^2 |f_i|^2.$$

The off-diagonal part is exactly $-\sigma x^* H x$, because H is Hermitian. This proves (7). $\square$

Corollary 3.2 (Exact two-sector variational formula). *Define*

$$\mathcal{E}_\sigma(f) = \sum_{\{i,j\}} r_i r_j \left[q_{ij} |f_i - \sigma \phi_{ij} f_j|^2 + (p_{ij} - q_{ij})(|f_i|^2 + |f_j|^2) \right]. \quad (8)$$

Then

$$\delta_\sigma(P, H) = \min_{f \neq 0} \frac{\mathcal{E}_\sigma(f)}{\sum_i r_i^2 |f_i|^2}. \quad (9)$$

In particular, both sector defects are nonnegative and $\|H\| \leq \rho(P)$.

Proof. Apply the Rayleigh principle to $\rho I - \sigma H$ and substitute $x_i = r_i f_i$. The substitution is bijective because every $r_i > 0$. $\square$

Remark 3.3 (Two independent sources of energy). The first term in (8) is phase or magnetic frustration. The second is amplitude-majorant slack. They are separately nonnegative; neither can cancel the other.

4 Complete saturation criterion

Theorem 4.1 (Connected Perron saturation). *Assume G_P is connected. The following are equivalent:*

- (i) $\|H\| = \rho(P)$;
- (ii) for some $\sigma \in \{\pm 1\}$, $\delta_\sigma(P, H) = 0$;
- (iii) for some $\sigma \in \{\pm 1\}$ and diagonal unitary $D = \text{diag}(d_i)$,

$$|H| = P, \quad H = \sigma D P D^*. \quad (10)$$

Proof. The equivalence of (i) and (ii) is (5). Suppose (ii) holds. A nonzero minimizer f in (9) has zero energy. If any edge has $q_{ij} < p_{ij}$, its amplitude term forces $f_i = f_j = 0$. Across any adjacent saturated edge, the phase term then forces the next vertex value to vanish; across any adjacent nonsaturated edge, the amplitude term does the same. Connectivity would give $f = 0$, a contradiction. Hence $q_{ij} = p_{ij}$ on every edge.

The remaining zero-energy equations are

$$f_i = \sigma \phi_{ij} f_j.$$

They imply equal modulus along every edge; connectivity gives $|f_i| = a > 0$ for all i . Put $d_i = f_i/a$. Then $\phi_{ij} = \sigma d_i \overline{d_j}$, which is (10).

Conversely, if (10) holds, H is unitarily similar to σP , so one of its extremal eigenvalues has modulus $\rho(P)$. $\square$

Corollary 4.2 (Pure nonnegative equality). *If $0 \leq A \leq P$, both matrices are symmetric, and G_P is connected, then*

$$\rho(A) = \rho(P) \iff A = P.$$

Proof. Apply theorem 4.1 with $H = A$. Since every nonzero edge phase equals 1, amplitude saturation is necessary and sufficient. $\square$

Corollary 4.3 (Exact amplitude–phase split). *For every Hermitian H dominated by P ,*

$$\boxed{\rho(P) - \|H\| = [\rho(P) - \rho(|H|)] + [\rho(|H|) - \|H\|].} \quad (11)$$

The first bracket is completion or amplitude slack. The second is phase-frustration slack. This algebraic identity is simple, but it prevents one source from being reported as evidence for the other.

5 Switching and cycle holonomy

Assume throughout this section that $P = |H|$ and G_P is connected.

Theorem 5.1 (Holonomy classification). *For a fixed $\sigma \in \{\pm 1\}$, the following are equivalent:*

(i) $H = \sigma D P D^*$ for a diagonal unitary D ;

(ii) there are phases d_i such that

$$\phi_{ij} = \sigma d_i \overline{d_j} \quad \text{on every edge};$$

(iii) every oriented cycle C satisfies

$$\text{Hol}_\phi(C) = \sigma^{|C|}, \quad \text{equivalently} \quad \text{Hol}_\sigma(C) = 1. \quad (12)$$

Proof. The first two statements are entrywise equivalent. Their phases telescopically give (12).

Conversely, fix a root and a spanning tree. Set the root phase arbitrarily and propagate d_i along the unique tree paths using $\phi_{ij} = \sigma d_i \overline{d_j}$. For a chord, its fundamental cycle condition says that the already assigned endpoint phases satisfy the same equation. Hence the formula holds on every edge. $\square$

Definition 5.2 (Balanced and antibalanced sectors). The phase pattern is *balanced* when the theorem holds for $\sigma = +1$, and *antibalanced* when it holds for $\sigma = -1$.

For a cycle of length ℓ , balanced means holonomy 1, whereas antibalanced means holonomy $(-1)^\ell$. On an odd graph the two conditions are distinct; on a bipartite graph they coincide after a bipartition sign switch.

Corollary 5.3 (Fundamental-cycle census). *Let T be any spanning tree of G_P . It is enough to test (12) on the*

$$|E| - |V| + 1$$

fundamental cycles formed by the chords $e \notin T$.

Proof. The spanning-tree construction in theorem 5.1 uses exactly those conditions. All other cycle holonomies are products of fundamental ones and their conjugates. $\square$

Remark 5.4 (Gauge normal form). A diagonal unitary can switch every tree-edge phase to $+1$. The remaining chord phase is, up to orientation conjugation, the holonomy of its fundamental cycle. Thus cycle rank, not edge count, is the number of independent scalar phase defects.

6 Trees cannot create scalar phase saving

Theorem 6.1 (Arbitrary phases on a tree). *Suppose $G_{|H|}$ is a tree and every tree edge is nonzero. Then a diagonal unitary D satisfies*

$$D^* H D = |H|.$$

Moreover the tree is bipartite, so another diagonal unitary switches H to $-|H|$. Consequently

$$\lambda_{\max}(H) = \rho(|H|), \quad \lambda_{\min}(H) = -\rho(|H|), \quad \|H\| = \rho(|H|). \quad (13)$$

Proof. There are no cycle constraints, so the root propagation in theorem 5.1 switches all phases to $+1$. Let $S = \text{diag}(s_i)$, where $s_i = +1$ on one bipartition class and -1 on the other. Then $S|H|S = -|H|$. Hence H is unitarily similar to both signs of $|H|$, and (13) follows. $\square$

Corollary 6.2 (Tree stop rule). *If every Perron-dominant connected component of the literal row graph $G_{|H|}$ is a tree, then the phase-frustration gap $\rho(|H|) - \|H\|$ is zero. Any strict gap relative to a larger majorant P is entirely amplitude/completion slack.*

Remark 6.3 (Not the TPC-78 tree). TPC-78 uses a bipartite graph whose vertices are analysis coordinates and completion variables. The present tree has physical Hermitian rows as vertices. One can be a forest while the other contains many cycles. No tree conclusion transfers without constructing the actual row matrix and its support.

7 Quantitative two-sector frustration bounds

Assume $P = |H|$ and G_P is connected. Define Perron conductances

$$c_{ij} = p_{ij}r_i r_j > 0. \quad (14)$$

Then [corollary 3.2](#) becomes

$$\delta_\sigma = \min_{f \neq 0} \frac{\sum_{\{i,j\}} c_{ij} |f_i - \sigma \phi_{ij} f_j|^2}{\sum_i r_i^2 |f_i|^2}. \quad (15)$$

Corollary 7.1 (Strictness criterion).

$$\delta_\sigma = 0 \iff \text{Hol}_\sigma(C) = 1 \text{ for every cycle } C.$$

Consequently,

$$\|H\| < \rho(P)$$

if and only if the phase pattern is neither balanced nor antibalanced.

Definition 7.2 (Cycle defect and unimodular frustration). For an oriented cycle C , set

$$\eta_\sigma(C) = |1 - \text{Hol}_\sigma(C)|.$$

Define

$$\mathfrak{F}_\sigma = \min_{\substack{z \in \mathbb{C}^V \\ |z_i|=1}} \sum_{\{i,j\}} c_{ij} |z_i - \sigma \phi_{ij} z_j|^2. \quad (16)$$

Theorem 7.3 (Fundamental-cycle upper certificate). For every spanning tree T ,

$$0 \leq \delta_\sigma \leq \mathfrak{F}_\sigma \leq \sum_{e \notin T} c_e \eta_\sigma(C_e)^2, \quad (17)$$

where C_e is the fundamental cycle of chord e .

Proof. Restricting (15) to unimodular $f = z$ makes its denominator $\sum_i r_i^2 = 1$, which gives $\delta_\sigma \leq \mathfrak{F}_\sigma$. Choose the vertex phases by propagating along T so that every tree edge has zero energy. On a chord e , the remaining mismatch is, up to a unit factor and orientation conjugation, $1 - \text{Hol}_\sigma(C_e)$. Evaluating this test phase proves the last inequality. $\square$

Small defects or defects carried by tiny c_e can therefore yield arbitrarily small strictness.

Theorem 7.4 (One-cycle resistance lower certificate). Fix a cycle C and a root $o \in C$ with $\eta_\sigma(C) > 0$. Set

$$R_C = \sum_{e \in C} c_e^{-1}.$$

For every $v \in V$, choose a path P_v from o to v and set

$$R_v = \sum_{e \in P_v} c_e^{-1}.$$

Then

$$\delta_\sigma \geq \left[\sum_{v \in V} r_v^2 \left(\frac{\sqrt{R_C}}{\eta_\sigma(C)} + \sqrt{R_v} \right)^2 \right]^{-1}. \quad (18)$$

In particular,

$$\delta_\sigma \geq \left(\frac{\sqrt{R_C}}{\eta_\sigma(C)} + \sqrt{\max_v R_v} \right)^{-2}. \quad (19)$$

Proof. Let $\mathcal{E}_\sigma(f)$ be the numerator in (15). Transport f once around C and telescope the edge errors. Weighted Cauchy–Schwarz gives

$$\eta_\sigma(C)|f_o| \leq \sqrt{R_C} \sqrt{\mathcal{E}_\sigma(f)}.$$

Transporting from o to v along P_v similarly gives

$$|f_v| \leq |f_o| + \sqrt{R_v} \sqrt{\mathcal{E}_\sigma(f)} \leq \left(\frac{\sqrt{R_C}}{\eta_\sigma(C)} + \sqrt{R_v} \right) \sqrt{\mathcal{E}_\sigma(f)}.$$

Multiply the squared inequalities by r_v^2 , sum, and insert the result into (15). Since $\sum_v r_v^2 = 1$, bounding every R_v by its maximum gives (19). $\square$

Corollary 7.5 (Certified operator saving). *If a defective cycle is supplied separately for each sector, with lower bounds $L_+ > 0$ and $L_- > 0$ from theorem 7.4, then*

$$\rho(P) - \|H\| \geq \min\{L_+, L_-\} > 0.$$

The cycles may be different. Finding a defect only in the positive sector is insufficient.

8 Exact flux spectrum on a cycle

Let P be the adjacency matrix of the unweighted cycle C_n , $n \geq 3$, so $\rho(P) = 2$. Let $|H| = P$, and suppose the total oriented holonomy is $e^{i\Theta}$.

Theorem 8.1 (Flux-cycle spectrum). *After switching and uniformly distributing the flux,*

$$\text{spec}(H) = \left\{ 2 \cos \left(\frac{\Theta + 2\pi k}{n} \right) : 0 \leq k < n \right\}. \quad (20)$$

Define

$$d_+(\Theta) = \text{dist}(\Theta, 2\pi\mathbb{Z}), \quad d_-(\Theta) = \text{dist}(\Theta, n\pi + 2\pi\mathbb{Z}),$$

with distances represented in $[0, \pi]$. Then

$$\delta_+ = 2 \left(1 - \cos \frac{d_+}{n} \right), \quad (21)$$

$$\delta_- = 2 \left(1 - \cos \frac{d_-}{n} \right), \quad (22)$$

$$2 - \|H\| = 2 \min_{\sigma \in \{+, -\}} \left(1 - \cos \frac{d_\sigma}{n} \right). \quad (23)$$

Proof. Switch all but one edge to phase 1, then conjugate by a diagonal unitary that distributes the remaining phase equally around the cycle. The Fourier vectors diagonalize the resulting twisted circulant and give (20). Maximizing the cosine gives (21). Multiplication of every edge by -1 changes total holonomy by $(-1)^n = e^{in\pi}$, which gives (22). Equation (5) gives the last formula. $\square$

Corollary 8.2 (Quadratic small-flux law). *For either sector,*

$$\frac{4d_\sigma^2}{\pi^2 n^2} \leq \delta_\sigma \leq \frac{d_\sigma^2}{n^2}. \quad (24)$$

Proof. Use

$$\frac{2t^2}{\pi^2} \leq 1 - \cos t \leq \frac{t^2}{2} \quad (|t| \leq \pi).$$

$\square$

Thus a nontrivial but shrinking flux can make every finite matrix strict while producing no uniform power-scale gap. For an even cycle, both sectors are balanced exactly at holonomy 1. For an odd cycle, operator-norm equality can occur at holonomy 1 through the positive sector or at holonomy -1 through the negative sector.

9 Reducible majorants and dominant components

Let

$$P = \bigoplus_{\alpha} P_{\alpha}$$

be the decomposition into connected row-support components. Entrywise domination forces the same block decomposition $H = \bigoplus_{\alpha} H_{\alpha}$. Set $\rho_{\alpha} = \rho(P_{\alpha})$ and

$$\mathcal{T}(P) = \{\alpha : \rho_{\alpha} = \rho(P)\}.$$

Theorem 9.1 (Reducible saturation).

$$\|H\| = \rho(P)$$

if and only if at least one $\alpha \in \mathcal{T}(P)$ satisfies, for some $\sigma \in \{\pm 1\}$ and diagonal unitary D_{α} ,

$$|H_{\alpha}| = P_{\alpha}, \quad H_{\alpha} = \sigma D_{\alpha} P_{\alpha} D_{\alpha}^*.$$

Other components may have arbitrary amplitude slack or frustration.

Proof. If $P = 0$, entrywise domination gives $H = 0$, and the assertion is immediate on any isolated component. Assume henceforth that $\rho(P) > 0$.

$$\|H\| = \max_{\alpha} \|H_{\alpha}\|, \quad \rho(P) = \max_{\alpha} \rho_{\alpha},$$

and $\|H_{\alpha}\| \leq \rho_{\alpha}$. Equality globally therefore requires and is implied by equality on a Perron-dominant component. Apply [theorem 4.1](#) there. $\square$

Corollary 9.2 (Exact component gap assembly).

$$\boxed{\rho(P) - \|H\| = \min_{\alpha} [\rho(P) - \rho_{\alpha} + \rho_{\alpha} - \|H_{\alpha}\|]} \quad (25)$$

Consequently, quantitative control on one frustrated dominant component is not sufficient if another dominant component still saturates. Conversely, frustration on a spectrally subdominant component cannot by itself change the global norm.

Corollary 9.3 (Reducible nonnegative equality). *If $0 \leq A \leq P$ are symmetric, then*

$$\rho(A) = \rho(P)$$

if and only if $A_{\alpha} = P_{\alpha}$ on at least one $\alpha \in \mathcal{T}(P)$.

10 The incidence graph and the row graph are not substitutes

The completion incidence graph of TPC-77/78 is bipartite:

$$G_C \subseteq I_{\text{analysis}} \sqcup J_{\text{variables}}.$$

It factorizes or solves the coefficient program $\min_{\theta} \|b + C\theta\|_1$. The present graph

$$G_{\text{row}}(P) \subseteq V_{\text{physical}} \times V_{\text{physical}}$$

controls the Perron eigenvector and scalar Hermitian phase transport.

Proposition 10.1 (Graph non-substitution). *Neither of the following implications holds without an additional literal construction:*

$$G_C \text{ is a forest} \implies G_{\text{row}}(P) \text{ is a forest,}$$

$$G_{\text{row}}(P) \text{ is a forest} \implies G_C \text{ is a forest.}$$

Proof. Take $C = I_4$. Its bipartite support is a matching, hence a forest, while an independently assembled row majorant may be the adjacency matrix of C_4 , whose row graph has a cycle.

Conversely, take a 2×2 matrix C with all four entries nonzero. Its support is $K_{2,2}$, which has a cycle, while a two-row Hermitian majorant with one nonzero off-diagonal entry has a tree row graph. These examples can share the same analysis-row labels; support of one matrix does not determine support of the other. $\square$

Remark 10.2 (Different phase transports). TPC-78 transports a dual charge through degree-two variable nodes of a bipartite tree. TPC-80 switches eigenvector phases along Hermitian row edges. Their equations, vertices, and obstruction spaces are different even when both happen to be one-dimensional.

The literal interface must therefore archive both graphs, with distinct names and hashes. A cycle counted before post- q entry aggregation is not a row holonomy cycle if its final Hermitian edge cancels to zero.

11 Sharp countermodels and scalar boundary

Example 11.1 (Arbitrary tree phases still saturate). *Put arbitrary unit phases and positive weights on the edges of any tree. Theorem 6.1 gives*

$$\|H\| = \rho(|H|).$$

Nonconstant phases alone are not evidence of cancellation.

Example 11.2 (The negative sector on a triangle).

$$H = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}$$

has total holonomy -1 and spectrum $\{-2, 1, 1\}$. It is not balanced, but it is antibalanced, so $\|H\| = 2 = \rho(|H|)$.

Example 11.3 (True frustration on an even cycle). *On C_4 , put phase -1 on one edge and $+1$ on the other three. Total holonomy is -1 , which fits neither sector because the cycle length is even. Theorem 8.1 gives*

$$\|H\| = \sqrt{2} < 2 = \rho(|H|).$$

Example 11.4 (Balanced phases with amplitude slack). *Let $H = (1 - \varepsilon)P$, where $0 < \varepsilon < 1$ and P is connected and nonzero. Every phase is balanced, but*

$$\|H\| = (1 - \varepsilon)\rho(P) < \rho(P).$$

This is completion slack, not phase frustration.

Example 11.5 (Frustration off the controlling component). *Take the direct sum of a balanced triangle and a frustrated C_4 ; both absolute components have Perron radius 2. The triangle still saturates the global norm. Finding one frustrated dominant component does not suffice when another dominant component is balanced.*

Example 11.6 (Finite strictness can vanish arbitrarily fast). *On a fixed cycle, take total flux $\Theta_X = X^{-100}$. Every finite X is strictly frustrated in the positive sector, but corollary 8.2 gives a defect $O(X^{-200})$. Strict inequality does not imply a fixed power useful to an arithmetic ledger.*

Remark 11.7 (Why scalar is essential). For matrix-valued edges, a cycle holonomy such as $\text{diag}(1, e^{i\vartheta}) \neq I$ can still have a common invariant first coordinate carrying a Perron eigenvector. Equality may require a common invariant section rather than trivial holonomy. For a general real-linear block $z \mapsto az + b\bar{z}$, there is no single edge phase. The scalar $U(1)$ classification must not be quoted for either setting.

12 Literal carrier specialization and claim boundary

Let $\tilde{H}_X$ be a finite scalar Hermitian carrier with zero diagonal, and let P_X be a finite real symmetric nonnegative zero-diagonal matrix. Earlier completion layers may supply such a majorant, including the most complete all-content or provenance-aware versions. Once the entrywise inequality

$$|\tilde{H}_X| \leq P_X$$

is checked on complete post-aggregation keys, the present equality theorem applies to this pair without further heuristic assumptions.

Proposition 12.1 (Literal Perron equality audit). *Under the finite scalar, symmetric/Hermitian, and zero-diagonal hypotheses above,*

$$\|\tilde{H}_X\| = \rho(P_X)$$

if and only if some Perron-dominant connected component of P_X simultaneously satisfies:

(i) *every edge majorant is tight,*

$$(P_X)_{mn} = |(\tilde{H}_X)_{mn}|;$$

(ii) *the exact entry phases are balanced or antibalanced on every fundamental row cycle.*

Proof. This is [theorems 5.1](#) and [9.1](#). □

The minimal archive records:

- (A1) fixed h_0 , scale X , complete row keys, and the common global raw normalizer;
- (A2) literal complex entries after every atomic/content sum and exact zero test;
- (A3) the chosen majorant and an entrywise slack table;
- (A4) connected components, their Perron radii and positive eigenvectors;
- (A5) a spanning forest, chord list, both sector holonomies, and Perron conductances;
- (A6) separate amplitude and phase gaps from [\(11\)](#);
- (A7) interval-verified eigenvalue or cycle certificates and data hashes.

No such growing literal carrier census is included here. Thus no claim is made that a dominant component contains a defective cycle, that both sectors have a uniform gap, or that the gap survives global normalization.

Progress and stop rules

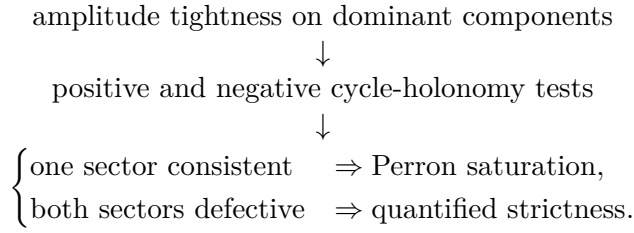
Level	Status
L0: spectral algebra	Exact for finite scalar Hermitian matrices.
L1: operator interface	Advanced by complete equality and weighted cycle-defect certificates.
L2: fixed- h_0 arithmetic	Open; no uniform growing-carrier or mass-normalized estimate is proved.

The branch stops as a phase-saving route if a Perron-dominant component of the exact post-aggregation graph $G_{|\tilde{H}_X|}$ is a tree or if any dominant component is balanced or antibalanced. Frustrating only one sector, thresholding an exact edge, or measuring a pre-aggregation atom phase is insufficient. Even a positive finite gap must be bounded uniformly on the complete growing family and connected to the physical scale.

The inherited exponent ledger permits total fixed-power loss strictly below $1/400$ and stops at loss at least $1/400$, including equality. This paper adds no exponent loss by itself and proves no exponent gain. A cycle gap cannot be entered as an arithmetic saving until a separate literal transfer theorem supplies its direction, normalization, and carrier coverage.

Conclusion

The Perron gap now has an exact decision procedure:



This closes the finite scalar Hermitian equality gate. The remaining question is arithmetic: whether the literal fixed-shift growing carrier contains sufficiently weighted, persistently defective dominant cycles.

References

- [1] Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*. SIAM, 1994. doi: 10.1137/1.9781611971262.
- [2] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, second edition, 2013. doi: 10.1017/CBO9781139020411.
- [3] Liang Wang. Primal–dual completion certificates for fixed-shift primitive entry kernels: Exact quotient gauges, obstruction witnesses, and perron aggregation, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-75-primal-dual-completion-certificates. Preprint.
- [4] Liang Wang. Exact forest reduction for sparse provenance completion: Matching rank, one-defect closed forms, and dual motif packings, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-78-exact-forest-completion-reduction. Preprint.